

Code with output:

```
#include <iostream>
#include <iomanip>
using namespace std;
// Definition of class HotelBill
class HotelBill
{
private:
// Data members (private for data security)
int tableNo, items; // Stores table number and number of items
string customerName; // Stores customer name
long long contact; // Stores customer contact number
float total, discount, payable; // Stores bill details
public:
// Member function to accept customer details
void getCustomerDetails()
{
// Accept table number
cout << "Enter Table Number: ";
cin >> tableNo;
// Clear input buffer before taking string input
cin.ignore();
// Accept customer name (including spaces)
cout << "Enter Customer Name: ";
getline(cin, customerName);
// Accept customer contact number
cout << "Enter Customer Contact Number: ";
cin >> contact;
}
// Member function to accept order details
void getOrderDetails()
{
int qty; // Quantity of each item
float price; // Price of each item
total = 0; // Initialize total bill amount
// Accept number of ordered items
cout << "\nEnter Number of Items Ordered: ";
cin >> items;
// Loop to accept quantity and price of each item
for (int i = 1; i <= items; i++)
{
cout << "\nItem " << i << " Quantity: ";
cin >> qty;
cout << "Item " << i << " Price: ";
```

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cin >> price;
// Calculate total amount
total += qty * price;
}
}
// Member function to calculate discount and final payable amount
void calculateBill()
{
// Apply discount based on total amount
if (total >= 2000)
discount = total * 0.20; // 20% discount
else if (total >= 1000)
discount = total * 0.10; // 10% discount
else
discount = 0; // No discount
// Calculate final payable amount
payable = total - discount;
}
// Member function to display the final bill
void displayBill()
{
cout << "\n----- HOTEL BILL ----- \n";
cout << "Table No : " << tableNo << endl;
cout << "Customer Name : " << customerName << endl;
cout << "Contact Number : " << contact << endl;
// Display bill amount with two decimal places
cout << fixed << setprecision(2);
cout << "Total Amount : Rs. " << total << endl;
cout << "Discount : Rs. " << discount << endl;
cout << "Payable Amount : Rs. " << payable << endl;
}
};
int main()
{
// Create an object of HotelBill class
HotelBill bill;
// Call member functions using object
bill.getCustomerDetails(); // Get customer information
bill.getOrderDetails(); // Get order details
bill.calculateBill(); // Calculate bill and discount
bill.displayBill();
// Display final bill
return 0; // End of program
}

```

OUTPUT :

Enter Table Number: 5

Enter Customer Name: Amit Sharma

Enter Customer Contact Number: 9876543210

Enter Number of Items Ordered: 3

Item 1 Quantity: 2

Item 1 Price: 250

Item 2 Quantity: 1

Item 2 Price: 600

Item 3 Quantity: 3

Item 3 Price: 300

----- HOTEL BILL -----

Table No : 5

Customer Name : Amit Sharma

Contact Number : 9876543210

Total Amount : Rs. 1850.00

Discount : Rs. 185.00

Payable Amount : Rs. 1665.00